Token embeddings from long-window co-occurrence as a Gaussian limit

Roadmap. Consider a large context window and forget token order, keeping only the count vector N. Since N is a sum of many weakly dependent one-hot token indicators, short-range dependence implies a multivariate central limit theorem: the centered fluctuations $(N - Lp)/\sqrt{L}$ become Gaussian. The leading-order statistics of large windows are therefore encoded by a single matrix, the long-run covariance Σ. A factorization $\Sigma = EE^\top$ then gives token vectors whose inner products reproduce these fluctuations. In this sense, embeddings arise as coordinates for the dominant collective modes of co-occurrence. Low-dimensional embeddings appear when Σ is well-approximated by its top eigenspaces; ordering and higher-order interactions enter only beyond this Gaussian level.

1 Setup

Let $\mathcal{V} = \{1, \dots, V\}$ be a finite vocabulary, and let $(w_t)_{t\in\mathbb{Z}}$ be a stationary token process. For a contiguous window of length L, define the count vector

$$N = (N_1, \dots, N_V)^\top, \qquad N_i = \#\{t \in \{1, \dots, L\} : w_t = i\},$$

so that

$$\sum_{i=1}^{V} N_i = L.$$

Introduce the one-hot process

$$Y_t \in \mathbb{R}^V, \qquad (Y_t)_i = \mathbf{1}_{\{w_t=i\}},$$

so that

$$N = \sum_{t=1}^{L} Y_t.$$

Let

$$p = \mathbb{E}[Y_t], \qquad p_i = \mathbb{P}(w_t = i).$$

We want to understand what notion of token embedding is naturally implied by the distribution of large context windows.

2 Long-window Gaussian limit

Assume the process is stationary with short-range dependence, in the sense that a multivariate CLT holds for additive observables and the covariance series is summable. Define the lag-τ covariance matrices

$$\Gamma(\tau) = \mathrm{Cov}(Y_0, Y_\tau),$$

and the long-run covariance

$$\Sigma = \sum_{\tau \in \mathbb{Z}} \Gamma(\tau).$$

Now define the centered, rescaled count fluctuations

$$X_L = \frac{N - Lp}{\sqrt{L}}.$$

Then

$$X_L \Rightarrow \mathcal{N}_H(0, \Sigma),$$

where the Gaussian is supported on the hyperplane

$$H = \{x \in \mathbb{R}^V : \mathbf{1}^\top x = 0\}.$$

The support restriction is forced by the exact identity $\mathbf{1}^\top N = L$, hence $\mathbf{1}^\top X_L = 0$. Accordingly,

$$\Sigma \mathbf{1} = 0, \qquad \mathbf{1}^\top \Sigma = 0.$$

Thus, at leading order in large windows, the probability law of token counts is Gaussian and completely determined by Σ.

3 Window covariances and the meaning of Σ

The covariance of the window counts satisfies

$$\mathrm{Cov}(N) = \sum_{s,t=1}^{L} \mathrm{Cov}(Y_s, Y_t) = \sum_{\tau=-(L-1)}^{L-1} (L - |\tau|)\Gamma(\tau).$$

Under summability of $\Gamma(\tau)$,

$$\mathrm{Cov}(N) = L\Sigma + o(L).$$

Equivalently,

$$\mathbb{E}[NN^\top] = L^2 pp^\top + L\Sigma + o(L).$$

So the raw second moment splits into a trivial mean piece $L^2 pp^\top$ and a nontrivial fluctuation piece $L\Sigma$. The latter is the meaningful second-order object for large-window statistics.

It is useful to separate

$$\Sigma = \Gamma(0) + \sum_{\tau \neq 0} \Gamma(\tau),$$

where

$$\Gamma(0) = \mathrm{diag}(p) - pp^\top.$$

The $\tau = 0$ term is present even for an i.i.d. process; it is just the multinomial baseline. Dependence across positions enters through the $\tau \neq 0$ terms. Depending on the application, one may work with the full Σ or with a dependence-only or normalized variant.

4 Embeddings from covariance factorization

Let

$$\Sigma = U_r \Lambda_r U_r^\top$$

be the eigendecomposition over the positive eigenspace, with

$$\Lambda_r = \operatorname{diag}(\lambda_1, \ldots, \lambda_r), \qquad \lambda_a > 0, \qquad r = \operatorname{rank}(\Sigma) \le V - 1.$$

Define

$$E = U_r \Lambda_r^{1/2}.$$

If $e_i^\top$ denotes the i-th row of E, then

$$\Sigma_{ij} = e_i \cdot e_j.$$

Hence the rows e_i define an exact second-order embedding of tokens: inner products between token vectors reproduce the long-run covariance of count fluctuations.

This is the canonical embedding implied by the Gaussian theory, up to the usual orthogonal freedom in factorizing a positive semidefinite matrix.

Low dimensionality, however, is not automatic. It emerges only if Σ is approximately low-rank. If the spectrum is concentrated, one may truncate to the top k modes:

$$\Sigma \approx U_k \Lambda_k U_k^\top = E_k E_k^\top, \qquad E_k = U_k \Lambda_k^{1/2}.$$

This gives the optimal rank-k approximation to Σ in the usual PCA sense.

5 Gaussian effective theory and additive features

Because Σ is singular, the Gaussian density is most naturally written on H:

$$P(X) \propto \exp\left(-\frac{1}{2} X^\top \Sigma^+ X\right), \qquad X \in H,$$

where Σ^+ is the Moore–Penrose pseudoinverse.

Let u_a be the eigenvectors corresponding to the positive eigenvalues λ_a. Define the whitened principal coordinates

$$q_a = \frac{u_a^\top X}{\sqrt{\lambda_a}}, \qquad a = 1, \ldots, r.$$

Then asymptotically

$$q \Rightarrow \mathcal{N}(0, I_r), \qquad P(q) \propto \exp\left(-\frac{1}{2} \sum_{a=1}^{r} q_a^2\right).$$

Since $X = (N - Lp)/\sqrt{L}$, each q_a can be written as an additive sum over token features:

$$q_a = \frac{1}{\sqrt{L}} \sum_{t=1}^{L} \psi_a(w_t),$$

with

$$\psi_a(i) = \frac{u_a(i) - \sum_j p_j u_a(j)}{\sqrt{\lambda_a}}.$$

So the Gaussian theory depends on the window only through additive sums of centered, whitened token features. This is the precise sense in which embeddings linearize the leading-order statistics of large contexts.

There are therefore two equivalent viewpoints:

1. the covariance embedding e_i, defined by $\Sigma = EE^\top$;
2. the whitened feature embedding $\psi(i)$, whose window sums are asymptotically standard Gaussian.

6 Relation to PCA, SVD, and standard co-occurrence methods

Empirically, Σ is estimated by taking many windows of length L, centering their count vectors, and computing the sample covariance. Diagonalizing that covariance is exactly PCA on window-count data. Equivalently, if one stacks centered window count vectors into a matrix, then the right singular vectors recover the principal eigenspaces of the empirical Σ.

So this framework gives a direct large-window justification for PCA/SVD-type embeddings.

It is important, however, not to conflate this with arbitrary co-occurrence or PMI matrices. Those are different statistics. In particular, for an i.i.d. token process, Σ is still nonzero because of the baseline term

$$\Gamma(0) = \mathrm{diag}(p) - pp^\top,$$

while nonzero-lag PMI vanishes. Thus diagonalizing Σ is not, in general, the same as factorizing PMI.

Conceptually, this construction is closer to PCA/LSA on large bag-of-words windows than to SGNS- or GloVe-style shifted-PMI factorizations.

7 Why the effective theory looks all-to-all

The Gaussian approximation for counts can be written as

$$P(N) \approx \exp\left[-\frac{1}{2L}(N - Lp)^\top \Sigma^+ (N - Lp)\right].$$

Substituting $N_i = \sum_{t=1}^{L} \mathbf{1}_{\{w_t = i\}}$ yields an effective interaction among all positions in the window:

$$-\frac{1}{2L} \sum_{s,t=1}^{L} K(w_s, w_t),$$

for a kernel K determined by Σ^+, up to one-body and constant terms.

So the coarse-grained description is indeed all-to-all across the window, with mean-field scaling $1/L$. But that all-to-all form is not what produces Gaussianity. Gaussianity comes from the CLT: the count vector is a sum of many weakly dependent contributions, and under short-range dependence only the second cumulant survives at leading order after $\sqrt{L}$ rescaling.

8 What this description captures — and what it misses

At the Gaussian level, the theory captures:

1. the full leading-order covariance of large-window bag-of-words fluctuations;
2. the dominant collective modes in which token counts co-vary;
3. the best low-rank second-order approximation obtained by spectral truncation.

What it does *not* capture includes:

1. ordering and positional structure inside the window;
2. higher cumulants of the count distribution;
3. nonlinear or multi-token interactions not reducible to second moments;
4. non-quadratic large-deviation behavior.

For short-range dependent processes, the m-th cumulant of the raw count vector typically scales as $O(L)$. Therefore the m-th cumulant of

$$X_L = \frac{N - Lp}{\sqrt{L}}$$

scales as

$$O\Big(L^{1-m/2}\Big).$$

So third-order corrections are $O(L^{-1/2})$, fourth-order corrections are $O(L^{-1})$, and so on. This is why the Gaussian theory is asymptotically exact in the central regime, and also why non-Gaussian structure can be organized systematically as a correction expansion.

9 Takeaway

In the large-window limit, the centered token-count vector satisfies a multivariate CLT and is therefore governed, to leading order, by a Gaussian law with covariance Σ, the long-run covariance of the one-hot token process. A factorization

$$\Sigma = EE^\top$$

then produces a canonical second-order embedding of tokens, and truncation to the leading eigenspaces gives the corresponding optimal low-rank representation.

This makes embeddings the natural first representation for large-context bag-of-words statistics: they are simply coordinates for the dominant Gaussian fluctuation modes. Everything beyond that — ordering, higher-order interactions, and genuinely non-Gaussian structure — lies in the correction to this leading theory.

10 Exact realization by a permutation-invariant architecture

It is useful to make the preceding Gaussian picture constructive. Suppose the window length L is fixed, and suppose the probability of a window depends only on its count vector

$$N = (N_1, \ldots, N_V)^\top, \qquad \sum_{i=1}^{V} N_i = L.$$

Assume further that this dependence is exactly quadratic in the counts, i.e. of Gaussian form on the count hyperplane. Then the corresponding probability law can be represented exactly by a permutation-invariant architecture of the form

$$\text{embedding lookup} \longrightarrow \text{sum pooling} \longrightarrow \text{quadratic energy head.}$$

In other words, no order-sensitive mechanism is needed: for this class of distributions, the sufficient statistic is simply the sum of token embeddings across the window.

10.1 Quadratic laws on count vectors

Let

$$H = \{x \in \mathbb{R}^V : \mathbf{1}^\top x = 0\}$$

be the count-fluctuation hyperplane. A general centered Gaussian-type energy on fixed-length windows has the form

$$\mathcal{E}(N) = \frac{1}{2}(N - \mu)^\top A(N - \mu),$$

where

$$\mathbf{1}^\top \mu = L, \qquad A \succeq 0 \text{ on } H, \qquad A\mathbf{1} = 0.$$

The corresponding permutation-invariant sequence model is

$$p(w_{1:L}) \propto \exp\bigl(-\mathcal{E}(N(w_{1:L}))\bigr).$$

Equivalently, one may view this as a discrete Gaussian on the lattice

$$\{N \in \mathbb{Z}^V_{\geq 0} : \mathbf{1}^\top N = L\}.$$

10.2 Sum-pooled covariance embeddings are sufficient

Now take the full, untruncated covariance-factorization embedding from the previous sections. Thus

$$\Sigma = EE^\top = U_r \Lambda_r U_r^\top, \qquad E = U_r \Lambda_r^{1/2},$$

with $r = \text{rank}(\Sigma)$, and let $e_i^\top$ denote the i-th row of E.

Given a window $w_{1:L}$, define the pooled embedding sum

$$s = \sum_{t=1}^{L} e_{w_t} = E^\top N.$$

This quantity is permutation invariant by construction.

The key point is that s contains exactly the same information as N along the support directions of the Gaussian theory. Indeed, on the support subspace

$$\text{Im}(\Sigma) \subseteq H,$$

the map

$$x \mapsto E^\top x$$

is injective. Hence any quadratic function of N restricted to the Gaussian support can be rewritten as a quadratic function of s.

More explicitly, if the model is supported on $\mu + \text{Im}(\Sigma)$, then there exist a positive semidefinite matrix M, a vector d, and a constant c such that

$$\mathcal{E}(N) = \frac{1}{2}(s - d)^\top M(s - d) + c.$$

Therefore the exact log-probability can be written as

$$\log p(w_{1:L}) = c' - \frac{1}{2}(s - d)^\top M(s - d), \qquad s = \sum_{t=1}^{L} e_{w_t}.$$

This already identifies the needed architecture: it is a *Deep Sets* model with a quadratic readout.

10.3 The exact architecture

A convenient parameterization is

$$x_t = e_{w_t} \in \mathbb{R}^r, \qquad s = \sum_{t=1}^{L} x_t, \qquad h = Ws + b,$$

followed by the scalar energy

$$\log p_\theta(w_{1:L}) = c - \frac{1}{2}\|h\|^2.$$

Equivalently,

$$\log p_\theta(w_{1:L}) = c - \frac{1}{2}(Ws + b)^\top (Ws + b).$$

This architecture is permutation invariant because it depends on the tokens only through the pooled sum s. It is also exactly expressive for the quadratic family above, provided the full embedding space is used.

Thus, for an exactly Gaussian law on token counts, the required architecture is:

$$\boxed{\text{token embedding lookup} \;\rightarrow\; \text{sum pooling} \;\rightarrow\; \text{linear map} \;\rightarrow\; \text{squared norm}}$$

and nothing more elaborate is needed.

10.4 Specialization to the Gaussian effective theory from Σ

For the specific Gaussian theory derived earlier,

$$p(w_{1:L}) \propto \exp\left[-\frac{1}{2L}(N - Lp)^\top \Sigma^+ (N - Lp)\right],$$

the quadratic head can be written in closed form in the covariance embedding coordinates.

Let

$$E = U_r \Lambda_r^{1/2}, \qquad s = E^\top N, \qquad \bar{s} = E^\top p.$$

Then

$$s - L\bar{s} = E^\top (N - Lp),$$

and the Gaussian energy becomes

$$\frac{1}{2L}(N - Lp)^\top \Sigma^+ (N - Lp) = \frac{1}{2L}(s - L\bar{s})^\top \Lambda_r^{-2} (s - L\bar{s}).$$

Hence an exact realization is obtained by choosing

$$h = L^{-1/2} \Lambda_r^{-1} (s - L\bar{s}),$$

so that

$$\log p(w_{1:L}) = c - \frac{1}{2}\|h\|^2.$$

Therefore the Gaussian effective theory from the previous sections is not merely approximable by such a model; it is exactly of this form.

10.5 Whitened additive features

The same construction becomes even simpler in the whitened basis introduced earlier. Define

$$\psi(i) = \big(\psi_a(i)\big)_{a=1}^{r}, \qquad \psi_a(i) = \frac{u_a(i) - \sum_j p_j u_a(j)}{\sqrt{\lambda_a}}.$$

Then

$$q = \frac{1}{\sqrt{L}} \sum_{t=1}^{L} \psi(w_t)$$

has asymptotically standard Gaussian statistics, and for the exact quadratic model above one has

$$\log p(w_{1:L}) = c - \frac{1}{2}\|q\|^2.$$

In that basis, the network is simply

$$\text{lookup whitened feature} \;\to\; \text{sum} \;\to\; \text{squared norm.}$$

10.6 Learning from data

If the embeddings are fixed in advance, then the only learnable part of the model is the quadratic head:

$$\log p_\theta(w_{1:L}) = c - \frac{1}{2}(Ws + b)^\top (Ws + b).$$

This is a finite-dimensional parametric family, and under correct specification standard maximum-likelihood arguments give consistency as the number of sampled windows grows.

If the embeddings are learned jointly with the head, the same architecture remains expressive enough, but the parameterization is no longer unique. There is a natural gauge freedom:

$$e_i \mapsto Re_i, \qquad W \mapsto WR^{-1},$$

for any invertible change of latent basis R. Thus the learned embedding coordinates are identifiable only up to such transformations, exactly as one would expect from a covariance factorization.

For the exact Gaussian setting, deeper set encoders, attention, recurrence, or positional structure are unnecessary. Those only become relevant once one moves beyond permutation-invariant quadratic laws.

10.7 Sequence probabilities versus count probabilities

A final bookkeeping point is worth making explicit. There are two distinct objects one might call a "Gaussian law on frequencies":

1. a Gaussian-type law on *sequences*, in which

$$p(w_{1:L}) \propto \exp\big(-\mathcal{E}(N(w_{1:L}))\big);$$

2. a Gaussian-type law on *count vectors*, in which

$$p_{\text{count}}(N) \propto \exp\big(-\mathcal{E}(N)\big).$$

These differ by the multinomial multiplicity factor

$$\#\{w_{1:L} : N(w_{1:L}) = N\} = \frac{L!}{\prod_i N_i!}.$$

Accordingly,

$$p_{\text{count}}(N) = \frac{L!}{\prod_i N_i!}\, p_{\text{seq}}(w_{1:L})$$

for any sequence $w_{1:L}$ with count vector N.

The architecture above directly models the permutation-invariant *sequence* probability. If one instead wishes to model the count-vector law itself, then this combinatorial factor must be included separately.

10.8 Takeaway

For a fixed-length window and a permutation-invariant probability law that is exactly quadratic in token counts, the correct architecture is a permutation-invariant energy-based model with sum pooling and a quadratic head. Using the full covariance-factorization embedding e_i, the pooled sum

$$s = \sum_{t=1}^{L} e_{w_t}$$

is already a sufficient statistic for the Gaussian support directions, and the exact probability takes the form

$$\log p(w_{1:L}) = c - \frac{1}{2}(Ws + b)^\top (Ws + b).$$

Thus the Gaussian count model is not merely compatible with embedding-based learning; it is exactly realized by the simplest possible set architecture.

10.9 Appendix: exact layer structure for the Gaussian count model

Fix a window length L, and consider the permutation-invariant sequence law

$$p(w_{1:L}) \propto \exp\left[-\frac{1}{2L}(N - Lp)^\top \Sigma^+ (N - Lp)\right],$$

where $N = (N_1, \ldots, N_V)^\top$ is the count vector of the window, p is the token-frequency vector, and

$$\Sigma = U_r \Lambda_r U_r^\top, \qquad E = U_r \Lambda_r^{1/2}.$$

Let $e_i^\top$ denote the i-th row of E, so that the covariance-factorization embedding of token i is $e_i \in \mathbb{R}^r$. Define also

$$\bar{e} := E^\top p \in \mathbb{R}^r.$$

The exact network realizing this law has the following layers.

Layer 0: token input. The input is a length-L token sequence

$$w_{1:L} \in \{1, \ldots, V\}^L.$$

Layer 1: embedding lookup. For each position t, look up the covariance embedding

$$x_t = e_{w_t} \in \mathbb{R}^r.$$

Layer 2: shared affine whitening. Apply the same affine map to every token:

$$z_t = \frac{1}{\sqrt{L}}\Lambda_r^{-1}(x_t - \bar{e}) = \frac{1}{\sqrt{L}}\Lambda_r^{-1}(e_{w_t} - \bar{e}).$$

Layer 3: permutation-invariant sum pooling. Sum the transformed token vectors across the window:

$$h = \sum_{t=1}^{L} z_t = \frac{1}{\sqrt{L}}\Lambda_r^{-1}\left(\sum_{t=1}^{L} e_{w_t} - L\bar{e}\right).$$

Since

$$\sum_{t=1}^{L} e_{w_t} = E^\top N,$$

this can be written as

$$h = \frac{1}{\sqrt{L}}\Lambda_r^{-1}E^\top(N - Lp).$$

Layer 4: quadratic energy head. Define the energy

$$\mathcal{E}(w_{1:L}) = \frac{1}{2}\|h\|^2.$$

The log-probability is then

$$\log p(w_{1:L}) = c - \mathcal{E}(w_{1:L})$$

for a normalizing constant c.

Verification. Using $E = U_r\Lambda_r^{1/2}$, we have

$$h = \frac{1}{\sqrt{L}}\Lambda_r^{-1}E^\top(N - Lp) = \frac{1}{\sqrt{L}}\Lambda_r^{-1/2}U_r^\top(N - Lp).$$

Therefore

$$\|h\|^2 = \frac{1}{L}(N - Lp)^\top U_r\Lambda_r^{-1}U_r^\top(N - Lp) = \frac{1}{L}(N - Lp)^\top\Sigma^+(N - Lp),$$

and hence

$$\frac{1}{2}\|h\|^2 = \frac{1}{2L}(N - Lp)^\top\Sigma^+(N - Lp).$$

So the architecture reproduces the target Gaussian law exactly.

Collapsed one-table form. Equivalently, one may absorb the affine whitening into a single lookup table

$$\phi(i) := \frac{1}{\sqrt{L}}\Lambda_r^{-1}(e_i - \bar{e}), \qquad i = 1, \ldots, V.$$

Then the model is simply

$$h = \sum_{t=1}^{L} \phi(w_t), \qquad \log p(w_{1:L}) = c - \frac{1}{2}\|h\|^2.$$

Remark. Thus the exact architecture is

$$\boxed{\text{token embedding lookup} \rightarrow \text{shared affine whitening} \rightarrow \text{sum pooling} \rightarrow \text{squared-norm head} \rightarrow \text{log-probability}}$$

with no attention, recurrence, or positional structure required. This is precisely a Deep Sets model with a quadratic readout.

Optional count-law correction. If one instead starts from a Gaussian law on count vectors $p_{\text{count}}(N)$, then the induced probability on ordered windows differs by the multiplicity factor

$$\frac{L!}{\prod_i N_i!}.$$

In that case,

$$\log p_{\text{seq}}(w_{1:L}) = \log p_{\text{count}}(N(w_{1:L})) - \log \frac{L!}{\prod_i N_i!}.$$

The architecture above gives the exact quadratic Gaussian energy; the multinomial multiplicity term must be added separately if one wants the exact sequence probability induced by a count-vector law.